Probability and Statistics

Chapter 6. Some Continuous Probability Distributions

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Chapter Overview

Distributions in Chapter 6

Distribution	Support	Typical Use / 용도
Continuous Uniform	$A \le x \le B$	같은 길이 구간이 같은 확률
Normal	$-\infty < x < \infty$	measurement error, natural variation
Gamma	$x > 0$	time until α Poisson events
Exponential	$x > 0$	time until first event, memoryless lifetime
Chi-Squared	$x > 0$	inference for variance, GOF, ANOVA
Beta	$0 < x < 1$	random proportion, reliability proportion
Lognormal	$x > 0$	positive skewed data; log becomes normal
Weibull	$x > 0$	lifetime/reliability with changing failure rate

Section

Continuous Uniform Distribution: pdf

Definition

A continuous random variable X has a **continuous uniform distribution** on $[A, B]$ if

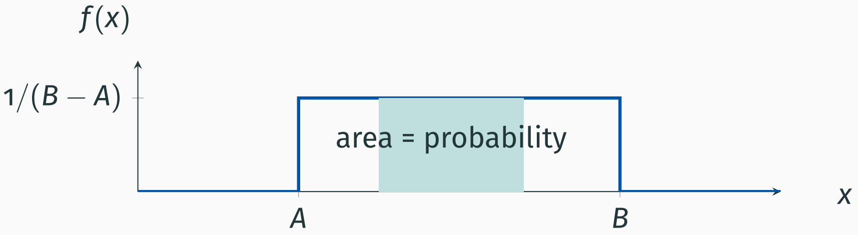
$$f(x; A, B) = \begin{cases} \dfrac{1}{B-A}, & A \leq x \leq B, \\ 0, & \text{elsewhere.} \end{cases}$$

We write $X \sim \text{Uniform}(A, B)$.

- Density is constant: "flat" or **rectangular distribution**.
- Probability over interval $[c, d] \subseteq [A, B]$:

$$\mathbb{P}(c \leq X \leq d) = \int_c^d \frac{1}{B-A} \mathrm{d}x = \frac{d-c}{B-A}.$$

Uniform Distribution: Geometry



Note

Uniform distribution에서 probability는 구간 길이만 보면 된다. 특정 점 하나의 확률은 항상 : $P(X = c) = 0$.

Theorem 6.1: Mean and Variance of Uniform

Theorem

If $X \sim \text{Uniform}(A, B)$, then

$$\mu = \mathbb{E}[X] = \frac{A+B}{2}, \qquad \sigma^2 = \text{Var}(X) = \frac{(B-A)^2}{12}.$$

Proof.

$$\mathbb{E}[X] = \int_A^B x \frac{1}{B-A} \mathrm{d}x = \frac{1}{B-A} \left[\frac{x^2}{2}\right]_A^B = \frac{B^2 - A^2}{2(B-A)} = \frac{A+B}{2}.$$

Also,

$$\mathbb{E}[X^2] = \int_A^B x^2 \frac{1}{B-A} \mathrm{d}x = \frac{B^3 - A^3}{3(B-A)} = \frac{A^2 + AB + B^2}{3}.$$

Thus

$$\text{Var}(X) = \mathbb{E}[X^2] - \{\mathbb{E}[X]\}^2 = \frac{A^2 + AB + B^2}{3} - \frac{(A+B)^2}{4} = \frac{(B-A)^2}{12}.$$

Example 6.1: Conference Room Reservation

어떤 회사의 큰 회의실은 최대 4시간까지 예약 가능하다. 회의 시간 X가 $[0, 4]$에서 uniform이라고 하자.

1. pdf를 구하여라.
2. 임의의 회의가 최소 3시간 이상 지속될 확률을 구하여라.

Example 6.1: Conference Room Reservation

Solution.

$$X \sim \text{Uniform}(0, 4), \qquad f(x) = \begin{cases} \frac{1}{4}, & 0 \le x \le 4, \\ 0, & \text{otherwise.} \end{cases}$$

$$\mathbb{P}(X \ge 3) = \int_3^4 \frac{1}{4} \mathrm{d}x = \frac{1}{4} = 0.25.$$

Note

회의 시간이 uniform이면, 3시간부터 4시간까지의 길이는 1이고 전체 길이는 4이므로 확률은 $1/4$.

Section

Normal Distribution: Motivation

- The **normal distribution** is the most important continuous distribution in statistics.
- It appears naturally in measurement errors, physical measurements, manufacturing variation, rainfall, scientific experiments, etc.
- The graph is called a **normal curve**, or bell-shaped curve.
- Also called **Gaussian distribution**.

Note

Normal distribution은 자연현상과 측정오차를 설명하는 대표적인 모델이며, 이후 통계적 추론(inference)의 핵심 기반임.

Normal Distribution: pdf

Definition

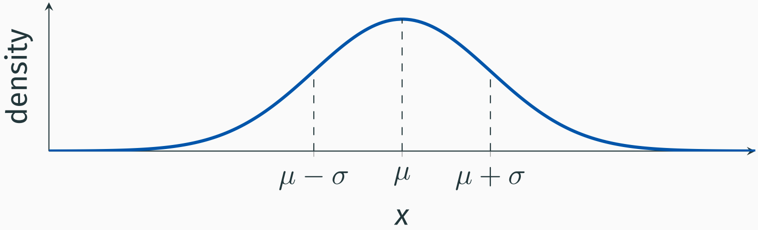
A continuous random variable X has a **normal distribution** with mean μ and variance σ^2 if

$$n(x;\mu,\sigma) = \frac{1}{\sqrt{2\pi}\sigma}\exp\left[-\frac{1}{2\sigma^2}(x-\mu)^2\right], \qquad -\infty < x < \infty.$$

We write $X \sim \mathcal{N}(\mu, \sigma^2)$.

- μ: center / location parameter.
- σ: spread / scale parameter.
- Total area under curve is 1.

Normal Curve: Shape



Note

μ가 바뀌면 curve가 좌우 이동하고, σ가 커지면 curve가 낮고 넓어짐.

Properties of Normal Curve

1. **Mode** occurs at $x = \mu$.
2. Curve is **symmetric** about $x = \mu$.
3. Inflection points occur at $x = \mu \pm \sigma$.
4. Curve approaches horizontal axis asymptotically.
5. Total area under curve is 1.

Note

Normal curve의 핵심은 대칭성이다. 평균보다 위/아래 같은 거리의 density가 동일.

Theorem 6.2: Mean and Variance of Normal

Theorem

If $X \sim \mathcal{N}(\mu, \sigma^2)$, then

$$\mathbb{E}[X] = \mu, \qquad \mathrm{Var}(X) = \sigma^2.$$

Proof. Let $Z = (X - \mu)/\sigma$. Then $dx = \sigma dz$ and

$$\mathbb{E}(X - \mu) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} z e^{-z^2/2} \mathrm{d}z = 0,$$

왜냐하면 integrand가 odd function이기 때문이다. Hence $\mathbb{E}[X] = \mu$.

Proof Continued: Variance of Normal

Proof.

$$\mathrm{Var}(X) = \mathbb{E}[(X-\mu)^2] = \frac{\sigma^2}{\sqrt{2\pi}} \int_{-\infty}^{\infty} z^2 e^{-z^2/2} \mathrm{d}z.$$

Integrate by parts:

$$u = z, \qquad dv = ze^{-z^2/2}\mathrm{d}z, \quad du = \mathrm{d}z, \quad v = -e^{-z^2/2}.$$

Then

$$\int_{-\infty}^{\infty} z^2 e^{-z^2/2} \mathrm{d}z = \left[-ze^{-z^2/2}\right]_{-\infty}^{\infty} + \int_{-\infty}^{\infty} e^{-z^2/2}\mathrm{d}z = \sqrt{2\pi}.$$

Therefore

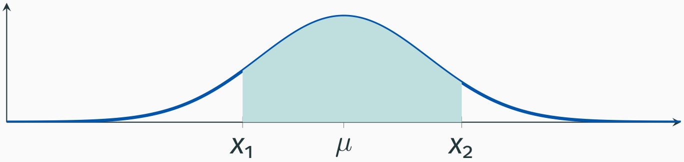
$$\mathrm{Var}(X) = \frac{\sigma^2}{\sqrt{2\pi}}\sqrt{2\pi} = \sigma^2.$$

Section

Probability as Area Under Normal Curve

For $X \sim \mathcal{N}(\mu, \sigma^2)$,

$$\mathbb{P}(x_1 < X < x_2) = \int_{x_1}^{x_2} n(x; \mu, \sigma)\mathrm{d}x.$$



Note

Normal probability는 density를 적분한 면적이다. 직접 적분하기 어렵기 때문에 standard normal table을 사용.

Standardization

Definition

If $X \sim \mathcal{N}(\mu, \sigma^2)$, then

$$Z = \frac{X - \mu}{\sigma} \sim \mathcal{N}(0, 1).$$

This is called the **standard normal distribution.**

$$\mathbb{P}(x_1 < X < x_2) = \mathbb{P}\left(\frac{x_1 - \mu}{\sigma} < Z < \frac{x_2 - \mu}{\sigma}\right).$$

Note

x값을 z값으로 바꾸는 이유는 모든 normal distribution을 하나의 표준정규분포표로 계산하기 위해서임.

Standard Normal Table: How to Use

- Table gives $\Phi(z) = \mathbb{P}(Z < z)$.
- Right-tail probability:

$$\mathbb{P}(Z > z) = 1 - \Phi(z).$$

- Between two values:

$$\mathbb{P}(a < Z < b) = \Phi(b) - \Phi(a).$$

- Reverse use: given area, find corresponding z value.

Note

왼쪽 누적확률(left area)을 기준으로 표를 읽는다. 오른쪽 확률은 반드시 $1-$을 해줌.

Example 6.2: Standard Normal Areas

$Z \sim \mathcal{N}(0, 1)$일 때 다음 area를 구하여라.

1. $P(Z > 1.84)$
2. $P(-1.97 < Z < 0.86)$

Solution.

$$P(Z > 1.84) = 1 - \Phi(1.84) = 1 - 0.9671 = 0.0329.$$

$$P(-1.97 < Z < 0.86) = \Phi(0.86) - \Phi(-1.97) = 0.8051 - 0.0244 = 0.7807.$$

Example 6.3: Find k from Normal Area

$Z \sim \mathcal{N}(0, 1)$일 때 k를 구하여라.

1. $P(Z > k) = 0.3015$
2. $P(k < Z < -0.18) = 0.4197$

Solution.

1. $P(Z < k) = 1 - 0.3015 = 0.6985$, so $k = 0.52$.
2. $\Phi(-0.18) = 0.4286$. Thus

$$P(Z < k) = 0.4286 - 0.4197 = 0.0089,$$

so $k = -2.37$.

Example 6.4: Normal Probability

$X \sim \mathcal{N}(50, 10^2)$일 때 $P(45 < X < 62)$를 구하여라.

Solution.

$$z_1 = \frac{45 - 50}{10} = -0.5, \qquad z_2 = \frac{62 - 50}{10} = 1.2.$$

$$P(45 < X < 62) = P(-0.5 < Z < 1.2) = \Phi(1.2) - \Phi(-0.5).$$

$$= 0.8849 - 0.3085 = 0.5764.$$

Note

Normal distribution 문제는 먼저 z-score로 바꾼 뒤 표준정규표를 읽음.

Example 6.5: Right Tail Probability

$X \sim \mathcal{N}(300, 50^2)$일 때 $P(X > 362)$를 구하여라.

Solution.

$$z = \frac{362 - 300}{50} = 1.24.$$

$$P(X > 362) = P(Z > 1.24) = 1 - \Phi(1.24) = 1 - 0.8925 = 0.1075.$$

Note

오른쪽 꼬리 확률은 $1 - \Phi(z)$.

Chebyshev vs Normal: Within 2 Standard Deviations

For any random variable, Chebyshev gives

$$P(\mu - 2\sigma < X < \mu + 2\sigma) \geq \frac{3}{4}.$$

If X is normal,

$$z_1 = -2, \qquad z_2 = 2.$$

Therefore

$$P(\mu - 2\sigma < X < \mu + 2\sigma) = P(-2 < Z < 2) = 0.9772 - 0.0228 = 0.9544.$$

Note

Normal assumption이 있으면 Chebyshev보다 훨씬 강한 결과를 얻음: 약 95.44%.

Using Normal Curve in Reverse

When area is given and x is unknown:

$$z = \frac{x - \mu}{\sigma} \implies x = \sigma z + \mu.$$

Note

역방향 문제는 **area** $\to$ z **value** $\to$ x **value** 순서로

Example 6.6: Reverse Normal Problem

$X \sim \mathcal{N}(40, 6^2)$일 때 다음 x값을 구하여라.

1. 왼쪽 면적이 45%인 x
2. 오른쪽 면적이 14%인 x

Solution.

1. $\Phi(z) = 0.45$이므로 $z = -0.13$.

$$x = 6(-0.13) + 40 = 39.22.$$

2. 오른쪽 면적 0.14이면 왼쪽 면적 0.86. $z = 1.08$.

$$x = 6(1.08) + 40 = 46.48.$$

Section

Normal Applications: General Procedure

1. Identify $X \sim \mathcal{N}(\mu, \sigma^2)$.
2. Draw/understand the desired area: left tail, right tail, or between.
3. Standardize: $z = (x - \mu)/\sigma$.
4. Use $\Phi(z)$ table.
5. Interpret probability in the original context.

Note

실제 문제에서는 “확률을 구하라”가 곧 “정규곡선 아래 어느 부분의 면적을 구하라”는 뜻.

Example 6.7: Battery Life

어떤 storage battery의 평균 수명은 3.0년, 표준편차는 0.5년이다. 수명이 normal 이라고 할 때, 한 battery가 2.3년 미만 지속될 확률을 구하여라.

Solution.

$$X \sim \mathcal{N}(3.0, 0.5^2), \qquad z = \frac{2.3 - 3.0}{0.5} = -1.4.$$

$$P(X < 2.3) = P(Z < -1.4) = 0.0808.$$

Example 6.8: Light Bulb Life

전구 수명이 평균 800시간, 표준편차 40시간인 normal distribution을 따른다. 전구가 778시간과 834시간 사이에 고장날 확률을 구하여라.

Solution.

$$z_1 = \frac{778 - 800}{40} = -0.55, \qquad z_2 = \frac{834 - 800}{40} = 0.85.$$

$$P(778 < X < 834) = P(-0.55 < Z < 0.85) = 0.8023 - 0.2912 = 0.5111.$$

Example 6.9: Ball Bearing Specification

Ball bearing diameter specification은 3.0 ± 0.01 cm이다. 공정에서 diameter는 $\mu = 3.0$, $\sigma = 0.005$인 normal distribution을 따른다. 평균적으로 몇 %가 scrap 되는가?

Solution. Acceptance interval: $2.99 < X < 3.01$.

$$z_1 = \frac{2.99 - 3.0}{0.005} = -2, \qquad z_2 = \frac{3.01 - 3.0}{0.005} = 2.$$

$$P(\text{scrap}) = P(Z < -2) + P(Z > 2) = 2(0.0228) = 0.0456.$$

따라서 약 4.56%가 폐기된다.

Example 6.10: Specification Width

어떤 dimension은 $N(1.50, 0.2^2)$를 따른다. Specification $1.50 \pm d$가 전체 측정값의 95%를 cover하도록 하는 d를 구하여라.

Solution.

$$P(-1.96 < Z < 1.96) = 0.95.$$

Upper specification:

$$1.96 = \frac{(1.50 + d) - 1.50}{0.2}.$$

$$d = 0.2(1.96) = 0.392.$$

Note

중앙 95% 구간은 approximately $\mu \pm 1.96\sigma$.

Example 6.11: Resistance Exceeding 43 Ohms

저항을 만드는 기계가 평균 40 ohms, 표준편차 2 ohms인 normal distribution을 따른다. 저항이 43 ohms를 초과하는 비율은?

Solution.

$$z = \frac{43 - 40}{2} = 1.5.$$

$$P(X > 43) = P(Z > 1.5) = 1 - 0.9332 = 0.0668.$$

따라서 약 6.68%이다.

Example 6.12: Measured to the Nearest Ohm

Example 6.11에서 resistance가 nearest ohm으로 측정된다면, 43 ohms를 초과한다고 기록될 비율은?

Solution. Nearest ohm 측정에서는 실제 값이 43.5를 넘어야 44 이상으로 기록된다.

$$z = \frac{43.5 - 40}{2} = 1.75.$$

$$P(X > 43.5) = P(Z > 1.75) = 1 - 0.9599 = 0.0401.$$

따라서 약 4.01%이다.

Note

Discrete로 기록되는 측정값에는 boundary correction이 필요.

Example 6.13: Curved Exam Grades

시험 평균은 74, 표준편차는 7이다. 성적이 normal distribution을 따르고 상위 12%에게 A를 준다면, 가능한 최저 A와 최고 B는?

Solution. A는 오른쪽 꼬리 0.12이므로 왼쪽 면적은 0.88.

$$\Phi(z) \approx 0.88 \Rightarrow z = 1.18.$$

$$x = 7(1.18) + 74 = 82.26.$$

따라서 최저 A는 83점, 최고 B는 82점으로 설정한다.

Example 6.14: Sixth Decile

Example 6.13에서 sixth decile D_6를 구하여라.

Solution. Sixth decile은 왼쪽 면적이 60%인 값이다.

$$\Phi(z) \approx 0.60 \Rightarrow z = 0.25.$$

$$x = 7(0.25) + 74 = 75.75.$$

따라서

$$D_6 = 75.75.$$

즉, 약 60%의 학생은 75.75점 이하이다.

Section

Why Approximate Binomial by Normal?

- Binomial probabilities can be computed exactly, but sums can be cumbersome for large n.
- A binomial histogram often becomes bell-shaped when n is large and p is not too close to 0 or 1.
- Then a normal distribution with the same mean and variance gives a good approximation.

$$X \sim \mathrm{Bin}(n, p), \qquad \mu = np, \qquad \sigma^2 = npq, \quad q = 1 - p.$$

Theorem 6.3: Limiting Normal Form

Theorem

If X is a binomial random variable with mean np and variance npq, then as $n \to \infty$,

$$Z = \frac{X - np}{\sqrt{npq}}$$

has limiting distribution $\mathcal{N}(0, 1)$.

Note

이것은 binomial distribution이 큰 표본에서는 normal distribution처럼 보인다는 뜻. 실제 계산에서는 $np \geq 5$ and $n(1 - p) \geq 5$를 흔한 기준으로 사용.

Continuity Correction

Binomial X is discrete, but normal Y is continuous.

$$P(X = x) \approx P(x - 0.5 < Y < x + 0.5).$$

$$P(a \leq X \leq b) \approx P(a - 0.5 < Y < b + 0.5).$$

Note

continuity correction: Discrete count를 continuous area로 바꿀 때는 ± 0.5를 붙임.

Normal Approximation Formula

Theorem

Let $X \sim \text{Bin}(n, p)$ and $q = 1 - p$. For large n,

$$X \approx Y \sim \mathcal{N}(np, npq).$$

Then

$$P(X \leq x) \approx P\left(Z \leq \frac{x + 0.5 - np}{\sqrt{npq}}\right).$$

Approximation is usually good if

$$np \geq 5, \qquad n(1-p) \geq 5.$$

Example: $b(x; 15, 0.4)$

For $X \sim \text{Bin}(15, 0.4)$,

$$\mu = np = 6, \qquad \sigma^2 = npq = 15(0.4)(0.6) = 3.6, \quad \sigma = 1.897.$$

Example point approximation:

$$P(X = 4) \approx P(3.5 < Y < 4.5).$$

$$z_1 = \frac{3.5 - 6}{1.897} = -1.32, \qquad z_2 = \frac{4.5 - 6}{1.897} = -0.79.$$

$$P(X = 4) \approx \Phi(-0.79) - \Phi(-1.32) = 0.2148 - 0.0934 = 0.1214.$$

Exact binomial value: 0.1268.

Binomial Sum Approximation Example

For $X \sim \text{Bin}(15, 0.4)$,

$$P(7 \leq X \leq 9) \approx P(6.5 < Y < 9.5).$$

$$z_1 = \frac{6.5 - 6}{1.897} = 0.26, \qquad z_2 = \frac{9.5 - 6}{1.897} = 1.85.$$

$$P(7 \leq X \leq 9) \approx \Phi(1.85) - \Phi(0.26) = 0.9678 - 0.6026 = 0.3652.$$

Exact binomial value: 0.3564.

Note

n이 커질수록 histogram이 normal curve와 더 비슷함.

Example 6.15: Rare Blood Disease

환자가 rare blood disease에서 회복할 확률이 0.4이다. 100명이 감염되었을 때, 생존자가 30명 미만일 확률은?

Solution.

$$X \sim \mathrm{Bin}(100, 0.4), \quad \mu = 40, \quad \sigma = \sqrt{100(0.4)(0.6)} = 4.899.$$

"Fewer than 30" means $X \leq 29$; continuity correction: 29.5.

$$z = \frac{29.5 - 40}{4.899} = -2.14.$$

$$P(X < 30) \approx P(Z < -2.14) = 0.0162.$$

Example 6.16: Multiple-Choice Guessing

200문항 중 학생이 전혀 모르는 80문항을 찍는다. 각 문항은 4지선다이고 정답은 1개이다. 찍어서 25개부터 30개까지 맞힐 확률은?

Solution.

$$X \sim \mathrm{Bin}(80, 1/4), \quad \mu = 20, \quad \sigma = \sqrt{80(1/4)(3/4)} = 3.873.$$

$$P(25 \le X \le 30) \approx P(24.5 < Y < 30.5).$$

$$z_1 = \frac{24.5 - 20}{3.873} = 1.16, \qquad z_2 = \frac{30.5 - 20}{3.873} = 2.71.$$

$$P \approx \Phi(2.71) - \Phi(1.16) = 0.9966 - 0.8770 = 0.1196.$$

Section

Why Gamma and Exponential?

- Normal distribution is powerful, but not enough for all engineering/science problems.
- Waiting time and reliability often require positive-valued distributions.
- **Exponential**: time to first event / time between events.
- **Gamma**: time until a specified number of events.

Note

Gamma와 Exponential은 “시간”이나 “수명”처럼 음수가 될 수 없는 random variable을 모델링하는 데 자주 사용됨.

Gamma Function

Definition

The **gamma function** is

$$\Gamma(\alpha) = \int_0^\infty x^{\alpha-1} e^{-x} \mathrm{d}x, \qquad \alpha > 0.$$

Important properties:

1. $\Gamma(\alpha) = (\alpha - 1)\Gamma(\alpha - 1)$ for $\alpha > 1$.
2. If n is a positive integer, $\Gamma(n) = (n - 1)!$.
3. $\Gamma(1) = 1$.
4. $\Gamma(1/2) = \sqrt{\pi}$.

Proof: Gamma Recursion Formula

Proof. Using integration by parts:

$$u = x^{\alpha-1}, \qquad dv = e^{-x}\mathrm{d}x, \qquad du = (\alpha-1)x^{\alpha-2}\mathrm{d}x, \qquad v = -e^{-x}.$$

Then

$$\Gamma(\alpha) = \left[-x^{\alpha-1}e^{-x}\right]_0^\infty + (\alpha-1)\int_0^\infty x^{\alpha-2}e^{-x}\mathrm{d}x.$$

Boundary term is 0, so

$$\Gamma(\alpha) = (\alpha-1)\Gamma(\alpha-1).$$

Repeated application gives $\Gamma(n) = (n-1)!$.

Gamma Distribution

Definition

A continuous random variable X has a **gamma distribution** with parameters $\alpha > 0$ and $\beta > 0$ if

$$f(x; \alpha, \beta) = \begin{cases} \frac{1}{\beta^\alpha \Gamma(\alpha)} x^{\alpha-1} e^{-x/\beta}, & x > 0, \\ 0, & \text{elsewhere.} \end{cases}$$

We write $X \sim \text{Gamma}(\alpha, \beta)$.

- α: shape parameter.
- β: scale parameter.

Exponential Distribution

Definition

The **exponential distribution** is a special gamma distribution with $\alpha = 1$.

$$f(x;\beta) = \begin{cases} \frac{1}{\beta} e^{-x/\beta}, & x > 0, \\ 0, & \text{elsewhere.} \end{cases}$$

We write $X \sim \text{Exponential}(\beta)$.

$$F(x) = P(X \le x) = 1 - e^{-x/\beta}, \qquad P(X > x) = e^{-x/\beta}.$$

Theorem 6.4 and Corollary 6.1

Theorem

If $X \sim \text{Gamma}(\alpha, \beta)$, then

$$\mathbb{E}[X] = \alpha\beta, \qquad \text{Var}(X) = \alpha\beta^2.$$

Theorem

If $X \sim \text{Exponential}(\beta)$, then

$$\mathbb{E}[X] = \beta, \qquad \text{Var}(X) = \beta^2.$$

Note

Exponential은 Gamma에서 $\alpha = 1$인 special case이므로 평균과 분산도 바로 구함.

Proof: Gamma Mean and Variance

Proof. If $X \sim \text{Gamma}(\alpha, \beta)$,

$$\mathbb{E}[X^r] = \int_0^\infty x^r \frac{1}{\beta^\alpha \Gamma(\alpha)} x^{\alpha-1} e^{-x/\beta} \mathrm{d}x.$$

Let $y = x/\beta$, so $x = \beta y$, $dx = \beta dy$.

$$\mathbb{E}[X^r] = \beta^r \frac{\Gamma(\alpha + r)}{\Gamma(\alpha)}.$$

Thus

$$\mathbb{E}[X] = \beta \frac{\Gamma(\alpha + 1)}{\Gamma(\alpha)} = \alpha\beta.$$

Also

$$\mathbb{E}[X^2] = \beta^2 \frac{\Gamma(\alpha + 2)}{\Gamma(\alpha)} = \beta^2 \alpha(\alpha + 1).$$

So

$$\text{Var}(X) = \mathbb{E}[X^2] - \mathbb{E}[X]^2 = \alpha(\alpha + 1)\beta^2 - \alpha^2\beta^2 = \alpha\beta^2.$$

Relationship to Poisson Process

For a Poisson process with rate λ events per unit time:

$$P(0 \text{ events in } [0, x]) = e^{-\lambda x}.$$

Let X be the waiting time until first event.

$$P(X > x) = e^{-\lambda x}.$$

So

$$F(x) = P(X \leq x) = 1 - e^{-\lambda x}, \qquad f(x) = \lambda e^{-\lambda x}.$$

This is exponential with

$$\lambda = \frac{1}{\beta}, \qquad \beta = \frac{1}{\lambda}.$$

Note

Poisson은 "일정 시간 동안 사건 개수", Exponential은 "다음 사건까지 기다리는 시간"을 모델링한다.

Memoryless Property

For $X \sim \text{Exponential}(\beta)$,

$$P(X \geq t_0 + t \mid X \geq t_0) = P(X \geq t).$$

Proof.

$$P(X \geq t_0 + t \mid X \geq t_0) = \frac{P(X \geq t_0 + t)}{P(X \geq t_0)} = \frac{e^{-(t_0+t)/\beta}}{e^{-t_0/\beta}} = e^{-t/\beta} = P(X \geq t).$$

Note

이미 오래 버틴 부품이라고 해서 앞으로 더 빨리 망가진다고 보지 않는 성질이다. Wear가 중요한 경우에는 exponential이 부적절할 수 있다.

Example 6.17: Components Still Functioning

어떤 component의 time to failure T는 exponential distribution이고 mean time to failure는 $\beta = 5$ years이다. 5개의 component를 서로 다른 시스템에 설치했을 때, 8년 후에도 최소 2개가 작동할 확률은?

Solution. 한 component가 8년 후에도 작동할 확률:

$$p = P(T > 8) = e^{-8/5} \approx 0.2.$$

Let $X =$ number of functioning components after 8 years.

$$X \sim \text{Bin}(5, 0.2).$$

$$P(X \geq 2) = 1 - P(X = 0) - P(X = 1) = 1 - 0.7373 = 0.2627.$$

Gamma as Waiting Time Until α Events

If events follow a Poisson process, then:

- time until first event $\Rightarrow$ exponential distribution.
- time until α-th event $\Rightarrow$ gamma distribution.

Note

Gamma distribution은 exponential waiting times를 여러 개 더한 것처럼 이해할 수 있다. 즉, 여러 사건이 발생할 때까지 기다리는 총 시간이다.

Example 6.18: Telephone Calls

전화 교환대에 평균 5 calls per minute으로 전화가 도착한다. 2번째 전화가 도착하기까지 걸리는 시간이 최대 1분일 확률은?

Solution. Poisson rate $\lambda = 5$, so $\beta = 1/5$. Time until 2 calls:

$$X \sim \text{Gamma}(\alpha = 2, \beta = 1/5).$$

$$P(X \leq 1) = \int_0^1 \frac{1}{\beta^2} x e^{-x/\beta} \mathrm{d}x = 25 \int_0^1 x e^{-5x} \mathrm{d}x.$$

$$= 1 - e^{-5}(1 + 5) \approx 0.96.$$

Incomplete Gamma Function

For gamma probabilities, define

$$F(x;\alpha) = \int_0^x \frac{y^{\alpha-1}e^{-y}}{\Gamma(\alpha)}\,\mathrm{d}y.$$

If $X \sim \text{Gamma}(\alpha,\beta)$, then using $y = x/\beta$,

$$P(X \le x_0) = F\left(\frac{x_0}{\beta};\alpha\right).$$

Note

Gamma cdf는 일반적으로 closed-form이 간단하지 않아 incomplete gamma table 또는 software를 사용한다.

Example 6.19: Rat Survival Time

어떤 toxicant dose에서 rat survival time X는 weeks 단위로 Gamma$(\alpha = 5, \beta = 10)$ 을 따른다. rat가 60주 이하로 생존할 확률은?

Solution.

$$P(X \le 60) = \int_0^{60} \frac{x^4 e^{-x/10}}{10^5 \Gamma(5)} \mathrm{d}x.$$

Let $y = x/10$. Then upper limit is 6:

$$P(X \le 60) = \int_0^{6} \frac{y^4 e^{-y}}{\Gamma(5)} \mathrm{d}y = F(6; 5).$$

Using incomplete gamma table:

$$F(6; 5) = 0.715.$$

Example 6.20: Customer Complaints

품질관리 개선 전에는 customer complaint 사이의 시간이 Gamma($\alpha = 2, \beta = 4$)를 따른다고 알려져 있다. 개선 후 첫 complaint까지 20개월이 걸렸다. 개선 효과가 있었는가?

Solution. 기존 조건에서 $X \geq 20$이 얼마나 rare한지 본다.

$$P(X \geq 20) = 1 - P(X \leq 20).$$

Let $y = x/4$, upper limit $20/4 = 5$.

$$P(X \geq 20) = 1 - F(5; 2) = 1 - 0.96 = 0.04.$$

Note

기존 모델에서 20개월 이상 complaint가 없는 것은 확률 4% 정도로 드문 사건이다. 따라서 개선이 효과적이었다고 볼 근거가 있다.

Example 6.21: Washing Machine Repair

Washing machine의 major repair 전 수명 Y가 다음 exponential density를 가진다.

$$f(y) = \frac{1}{4}e^{-y/4}, \qquad y \geq 0.$$

$P(Y > 6)$와 첫 1년 안에 major repair가 필요한 확률을 구하여라.

Solution. Here $\beta = 4$.

$$F(y) = 1 - e^{-y/4}.$$

$$P(Y > 6) = e^{-6/4} = e^{-3/2} = 0.2231.$$

$$P(Y < 1) = 1 - e^{-1/4} = 1 - 0.779 = 0.221.$$

Section

Chi-Squared Distribution

Definition

A continuous random variable X has a **chi-squared distribution** with v degrees of freedom if

$$f(x; v) = \begin{cases} \dfrac{1}{2^{v/2}\Gamma(v/2)} x^{v/2-1} e^{-x/2}, & x > 0, \\ 0, & \text{elsewhere.} \end{cases}$$

We write $X \sim \chi^2_v$.

This is a gamma distribution with

$$\alpha = \frac{v}{2}, \qquad \beta = 2.$$

Theorem 6.5: Mean and Variance of Chi-Squared

Theorem

If $X \sim \chi^2_\nu$, then

$$\mathbb{E}[X] = \nu, \qquad \mathrm{Var}(X) = 2\nu.$$

Proof. Since χ^2_ν is $\mathrm{Gamma}(\alpha = \nu/2, \beta = 2)$,

$$\mathbb{E}[X] = \alpha\beta = \frac{\nu}{2} \cdot 2 = \nu,$$

$$\mathrm{Var}(X) = \alpha\beta^2 = \frac{\nu}{2} \cdot 4 = 2\nu.$$

Note

Chi-squared distribution은 나중에 variance inference, goodness-of-fit test, ANOVA 등에 핵심적으로 사용된다.

Section

Beta Function

Definition

The **beta function** is

$$B(\alpha, \beta) = \int_0^1 x^{\alpha-1}(1-x)^{\beta-1}\mathrm{d}x = \frac{\Gamma(\alpha)\Gamma(\beta)}{\Gamma(\alpha+\beta)}, \qquad \alpha, \beta > 0.$$

Beta Distribution

Definition

A continuous random variable X has a **beta distribution** with parameters $\alpha > 0$ and $\beta > 0$ if

$$f(x) = \begin{cases} \frac{1}{B(\alpha, \beta)} x^{\alpha-1}(1-x)^{\beta-1}, & 0 < x < 1, \\ 0, & \text{elsewhere.} \end{cases}$$

We write $X \sim \text{Beta}(\alpha, \beta)$.

- Models random proportions.
- $\text{Uniform}(0, 1)$ is $\text{Beta}(1, 1)$.

Theorem 6.6: Mean and Variance of Beta

Theorem

If $X \sim \text{Beta}(\alpha, \beta)$, then

$$\mathbb{E}[X] = \frac{\alpha}{\alpha + \beta}, \qquad \text{Var}(X) = \frac{\alpha\beta}{(\alpha + \beta)^2(\alpha + \beta + 1)}.$$

For $\text{Uniform}(0, 1) = \text{Beta}(1, 1)$,

$$\mathbb{E}[X] = \frac{1}{2}, \qquad \text{Var}(X) = \frac{1}{12}.$$

Proof: Beta Mean and Variance

Proof. For $X \sim \text{Beta}(\alpha, \beta)$,

$$\mathbb{E}[X^r] = \frac{1}{B(\alpha,\beta)} \int_0^1 x^{\alpha+r-1}(1-x)^{\beta-1}\mathrm{d}x = \frac{B(\alpha+r,\beta)}{B(\alpha,\beta)}.$$

Using $B(\alpha,\beta) = \Gamma(\alpha)\Gamma(\beta)/\Gamma(\alpha+\beta)$,

$$\mathbb{E}[X] = \frac{\Gamma(\alpha+1)\Gamma(\alpha+\beta)}{\Gamma(\alpha)\Gamma(\alpha+\beta+1)} = \frac{\alpha}{\alpha+\beta}.$$

Also

$$\mathbb{E}[X^2] = \frac{\alpha(\alpha+1)}{(\alpha+\beta)(\alpha+\beta+1)}.$$

Therefore

$$\mathrm{Var}(X) = \mathbb{E}[X^2] - \mathbb{E}[X]^2 = \frac{\alpha\beta}{(\alpha+\beta)^2(\alpha+\beta+1)}.$$

Section

Lognormal Distribution

Definition

A continuous random variable X has a **lognormal distribution** if

$$Y = \ln(X)$$

has a normal distribution with mean μ and standard deviation σ. Then

$$f(x;\mu,\sigma) = \begin{cases} \dfrac{1}{\sqrt{2\pi}\sigma x} \exp\left[-\dfrac{1}{2\sigma^2}(\ln x - \mu)^2\right], & x > 0, \\ 0, & x \le 0. \end{cases}$$

Note

Lognormal은 원자료 X가 아니라 $\ln X$가 normal distribution을 따른다는 뜻이다.

Theorem 6.7: Mean and Variance of Lognormal

Theorem

If X is lognormal with $\ln X \sim \mathcal{N}(\mu, \sigma^2)$, then

$$\mathbb{E}[X] = e^{\mu+\sigma^2/2},$$

$$\mathrm{Var}(X) = e^{2\mu+\sigma^2}\left(e^{\sigma^2} - 1\right).$$

Proof: Lognormal Mean and Variance

Proof. Let $Y = \ln X \sim \mathcal{N}(\mu, \sigma^2)$, so $X = e^Y$. For normal Y, moment generating function is

$$M_Y(t) = \mathbb{E}[e^{tY}] = \exp\left(\mu t + \frac{\sigma^2 t^2}{2}\right).$$

Thus

$$\mathbb{E}[X] = \mathbb{E}[e^Y] = M_Y(1) = e^{\mu+\sigma^2/2}.$$

Also

$$\mathbb{E}[X^2] = \mathbb{E}[e^{2Y}] = M_Y(2) = e^{2\mu+2\sigma^2}.$$

Therefore

$$\mathrm{Var}(X) = e^{2\mu+2\sigma^2} - e^{2\mu+\sigma^2} = e^{2\mu+\sigma^2}(e^{\sigma^2} - 1).$$

Example 6.22: Pollutant Concentration

어떤 pollutant concentration X가 lognormal distribution을 따르고, $\ln X \sim N(3.2, 1^2)$이다. 농도가 8 ppm을 초과할 확률은?

Solution.

$$P(X > 8) = 1 - P(X \le 8).$$

Since $\ln X \sim N(3.2, 1^2)$,

$$P(X \le 8) = P(\ln X \le \ln 8) = \Phi\left(\frac{\ln 8 - 3.2}{1}\right).$$

$$\frac{\ln 8 - 3.2}{1} \approx -1.12, \quad \Phi(-1.12) = 0.1314.$$

Thus

$$P(X > 8) = 1 - 0.1314 = 0.8686.$$

Example 6.23: Lognormal Percentile

기관차 electronic control의 life X는 thousand miles 단위로 lognormal이고, $\ln X \sim N(5.149, 0.737^2)$이다. 5th percentile을 구하여라.

Solution. 5th percentile means $P(X < x) = 0.05$.

$$P(Z < -1.645) = 0.05.$$

Thus

$$\ln x = 5.149 + 0.737(-1.645) = 3.937.$$

$$x = e^{3.937} = 51.265.$$

즉, 약 51.265 thousand miles, 곧 51,265 miles이다.

Section

Weibull Distribution: Motivation

- Used extensively in reliability and life-testing.
- More flexible than exponential distribution.
- Can model increasing, decreasing, or constant failure rates.

Note

Exponential은 failure rate가 constant인 특별한 경우이다. Weibull은 component가 시간이 지나며 약해지거나 강해지는 상황도 표현할 수 있다.

Weibull Distribution: pdf

Definition

A continuous random variable X has a **Weibull distribution** with parameters $\alpha > 0$ and $\beta > 0$ if

$$f(x;\alpha,\beta) = \begin{cases} \alpha\beta x^{\beta-1} e^{-\alpha x^{\beta}}, & x > 0, \\ 0, & \text{elsewhere.} \end{cases}$$

If $\beta = 1$, Weibull reduces to exponential with rate α.

Weibull CDF

Theorem

For $X \sim \text{Weibull}(\alpha, \beta)$,

$$F(x) = P(X \leq x) = 1 - e^{-\alpha x^{\beta}}, \qquad x \geq 0.$$

Proof.

$$F(x) = \int_0^x \alpha\beta t^{\beta-1} e^{-\alpha t^{\beta}} \mathrm{d}t.$$

Let $u = \alpha t^{\beta}$, then $du = \alpha\beta t^{\beta-1}\mathrm{d}t$.

$$F(x) = \int_0^{\alpha x^{\beta}} e^{-u} \mathrm{d}u = 1 - e^{-\alpha x^{\beta}}.$$

Theorem 6.8: Weibull Mean and Variance

Theorem

If $X \sim \text{Weibull}(\alpha, \beta)$, then

$$\mathbb{E}[X] = \alpha^{-1/\beta}\Gamma\left(1 + \frac{1}{\beta}\right),$$

$$\text{Var}(X) = \alpha^{-2/\beta}\left[\Gamma\left(1 + \frac{2}{\beta}\right) - \left\{\Gamma\left(1 + \frac{1}{\beta}\right)\right\}^2\right].$$

Proof: Weibull Moments

Proof. For $r > 0$,

$$\mathbb{E}[X^r] = \int_0^\infty x^r \alpha\beta x^{\beta-1} e^{-\alpha x^\beta}\,\mathrm{d}x.$$

Let $u = \alpha x^\beta$, so $x = (u/\alpha)^{1/\beta}$ and $du = \alpha\beta x^{\beta-1}\mathrm{d}x$.

$$\mathbb{E}[X^r] = \int_0^\infty \left(\frac{u}{\alpha}\right)^{r/\beta} e^{-u}\mathrm{d}u = \alpha^{-r/\beta}\Gamma\left(1+\frac{r}{\beta}\right).$$

Thus

$$\mathbb{E}[X] = \alpha^{-1/\beta}\Gamma\left(1+\frac{1}{\beta}\right), \quad \mathbb{E}[X^2] = \alpha^{-2/\beta}\Gamma\left(1+\frac{2}{\beta}\right).$$

Variance follows from $\mathrm{Var}(X) = \mathbb{E}[X^2] - \{\mathbb{E}[X]\}^2$.

Example 6.24: Weibull Failure Probability

Machine shop item의 lifetime X가 Weibull distribution을 따르고 $\alpha = 0.01$, $\beta = 2$ 이다. 8시간 전에 fail할 확률은?

Solution.

$$P(X < 8) = F(8) = 1 - e^{-0.01(8)^2} = 1 - e^{-0.64}.$$

$$e^{-0.64} \approx 0.527, \qquad P(X < 8) = 1 - 0.527 = 0.473.$$

Reliability and Failure Rate

Definition

Reliability at time t:

$$R(t) = P(T > t) = 1 - F(t).$$

Failure rate / hazard rate:

$$Z(t) = \frac{f(t)}{1 - F(t)} = \frac{f(t)}{R(t)}.$$

For Weibull,

$$f(t) = \alpha\beta t^{\beta-1} e^{-\alpha t^{\beta}}, \qquad 1 - F(t) = e^{-\alpha t^{\beta}}.$$

Thus

$$Z(t) = \alpha\beta t^{\beta-1}.$$

Interpreting Weibull Failure Rate

- If $\beta = 1$, then $Z(t) = \alpha$: constant failure rate. Exponential case.
- If $\beta > 1$, then $Z(t)$ increases with t: component wears over time.
- If $\beta < 1$, then $Z(t)$ decreases with t: component strengthens/hardens over time.

Note

Weibull의 가장 중요한 장점은 failure rate의 시간 변화 패턴을 parameter β 하나로 해석할 수 있다는 점이다.

Section

Potential Misconceptions and Hazards

- Do not assume normality automatically.
- Normal-based inference is reliable only when the normality assumption is reasonable.
- Distribution choice matters: exponential vs Weibull vs gamma can lead to very different reliability predictions.
- Parameters are often unknown in real applications and must be estimated from data.

Note

가장 큰 위험은 “무조건 normal distribution이라고 가정하는 것”이다. 실제 데이터에서는 그래프, goodness-of-fit test, domain knowledge로 분포 가정을 확인해야 한다.

Summary Table

Distribution	pdf core	Mean	Variance
Uniform(A, B)	$1/(B-A)$	$(A+B)/2$	$(B-A)^2/12$
Normal(μ, σ^2)	$e^{-(x-\mu)^2/(2\sigma^2)}$	μ	σ^2
Gamma(α, β)	$x^{\alpha-1}e^{-x/\beta}$	$\alpha\beta$	$\alpha\beta^2$
Exp(β)	$e^{-x/\beta}/\beta$	β	β^2
χ^2_v	Gamma$(v/2, 2)$	v	$2v$
Beta(α, β)	$x^{\alpha-1}(1-x)^{\beta-1}$	$\alpha/(\alpha+\beta)$	$\frac{\alpha\beta}{(\alpha+\beta)^2(\alpha+\beta+1)}$
Lognormal	$\ln X \sim N(\mu, \sigma^2)$	$e^{\mu+\sigma^2/2}$	$e^{2\mu+\sigma^2}(e^{\sigma^2}-1)$
Weibull(α, β)	$\alpha\beta x^{\beta-1}e^{-\alpha x^\beta}$	$\alpha^{-1/\beta}\Gamma(1+1/\beta)$	see theorem

Summary Notes

Note

1. Continuous probability는 density가 아니라 **area**이다.
2. Normal 문제는 거의 항상 **standardization**으로 시작한다.
3. Binomial을 normal로 근사할 때는 **continuity correction**을 사용한다.
4. Exponential은 **memoryless**이고, Weibull은 failure rate가 변할 수 있다.
5. Gamma, Exponential, Chi-squared는 서로 밀접히 연결되어 있다.
6. Beta는 proportion, Lognormal은 positive skewed data에 적합하다.

Appendix: Quick Formula Bank i

Uniform

$$f(x) = \frac{1}{B-A}, \quad \mathbb{E}[X] = \frac{A+B}{2}, \quad \mathrm{Var}(X) = \frac{(B-A)^2}{12}.$$

Normal

$$f(x) = \frac{1}{\sqrt{2\pi}\sigma} e^{-(x-\mu)^2/(2\sigma^2)}, \quad Z = \frac{X-\mu}{\sigma}.$$

Binomial Normal Approximation

$$X \sim \mathrm{Bin}(n, p), \quad X \approx N(np, npq), \quad P(X \le x) \approx \Phi\left(\frac{x + 0.5 - np}{\sqrt{npq}}\right).$$

Appendix: Quick Formula Bank ii

Gamma

$$f(x) = \frac{1}{\beta^\alpha \Gamma(\alpha)} x^{\alpha-1} e^{-x/\beta}, \quad \mathbb{E}[X] = \alpha\beta, \quad \mathrm{Var}(X) = \alpha\beta^2.$$

Exponential

$$f(x) = \frac{1}{\beta} e^{-x/\beta}, \quad F(x) = 1 - e^{-x/\beta}, \quad P(X > x) = e^{-x/\beta}.$$

Chi-squared

$$\chi^2_v = \mathrm{Gamma}(v/2, 2), \quad \mathbb{E}[X] = v, \quad \mathrm{Var}(X) = 2v.$$

Appendix: Quick Formula Bank iii

Beta

$$f(x) = \frac{1}{B(\alpha,\beta)} x^{\alpha-1}(1-x)^{\beta-1}, \quad 0 < x < 1.$$

$$\mathbb{E}[X] = \frac{\alpha}{\alpha+\beta}, \quad \mathrm{Var}(X) = \frac{\alpha\beta}{(\alpha+\beta)^2(\alpha+\beta+1)}.$$

Lognormal

$$\ln X \sim N(\mu, \sigma^2), \quad \mathbb{E}[X] = e^{\mu+\sigma^2/2}.$$

Weibull

$$f(x) = \alpha\beta x^{\beta-1} e^{-\alpha x^{\beta}}, \quad F(x) = 1 - e^{-\alpha x^{\beta}}, \quad Z(t) = \alpha\beta t^{\beta-1}.$$